

Bank Customer Churn Analysis

Total Customers

10000

Total Customers

Churned Customers

2037

Churned Customers

Retained Customers

7963

Retained Customers

Churn Rate

20.4%

Churn Rate

Geography

☐ France

☐ Germany

☐ Spain

Active Status

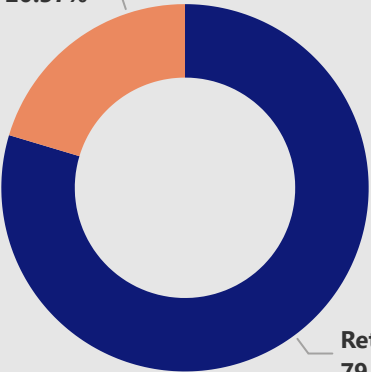
☐ Select all

☐ Active

☐ Inactive

Customer Churn Split

Churned 20.37%



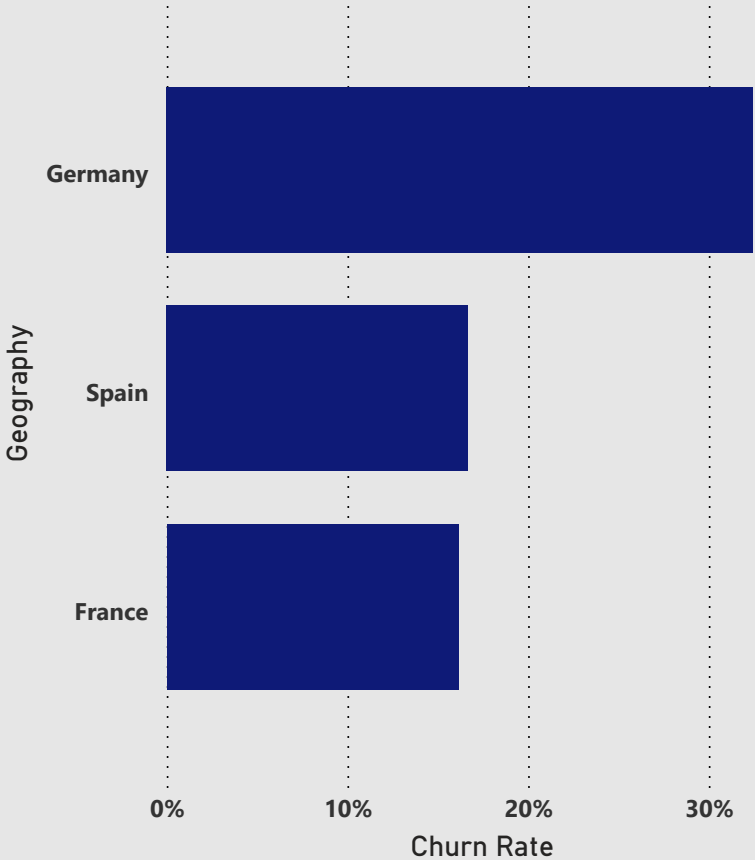
Retained
79.63%

Churn Status

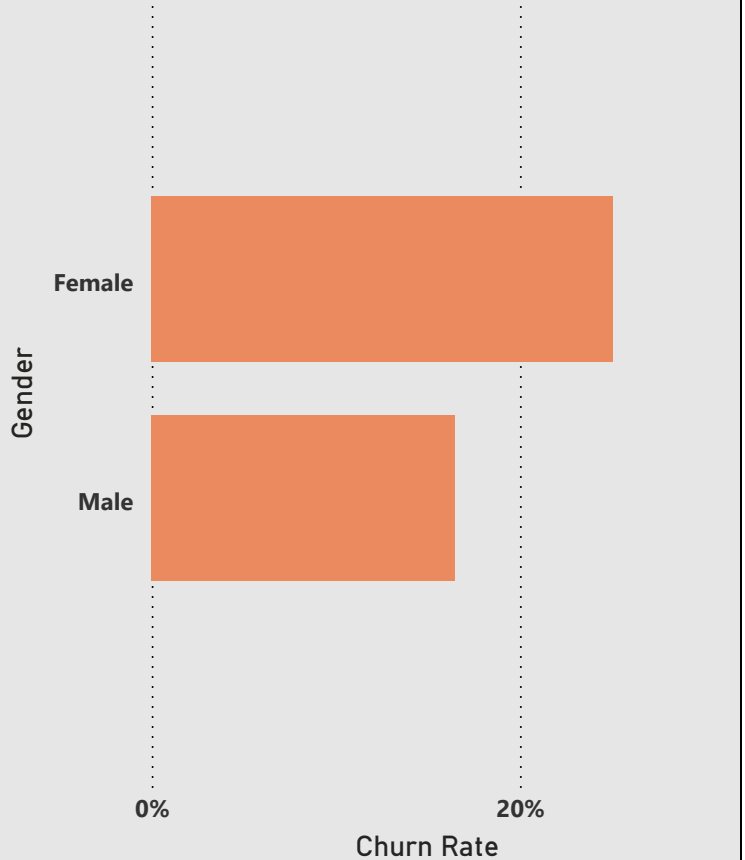
● Retained

● Churned

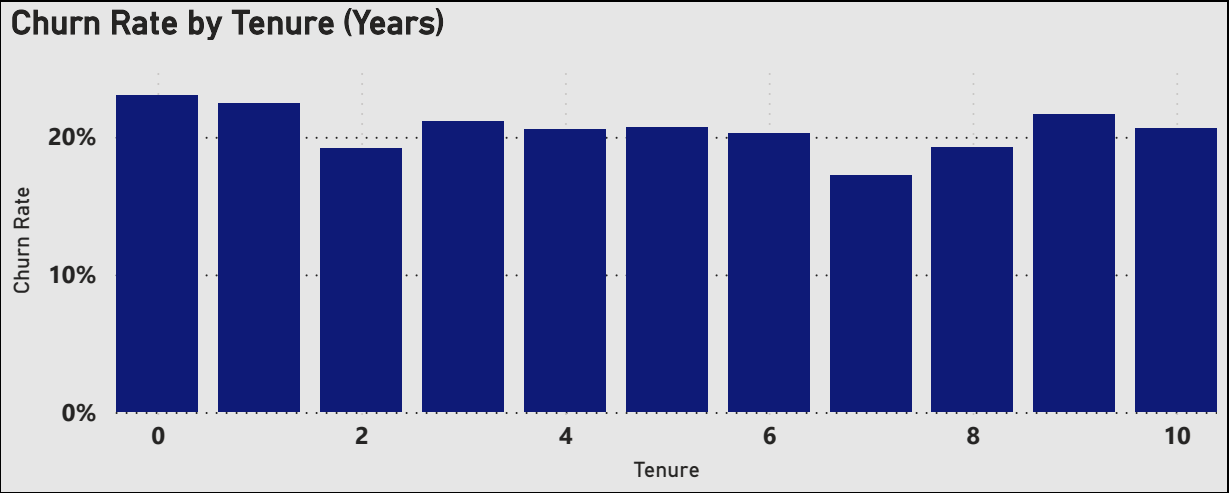
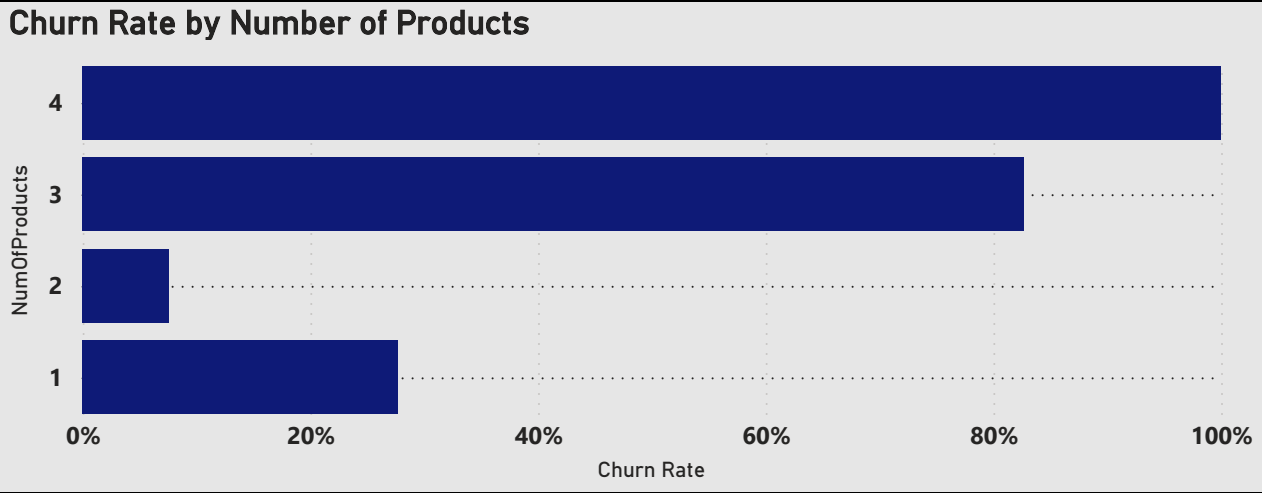
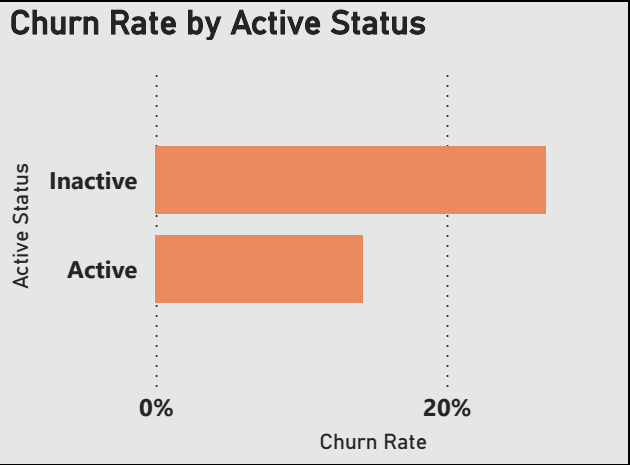
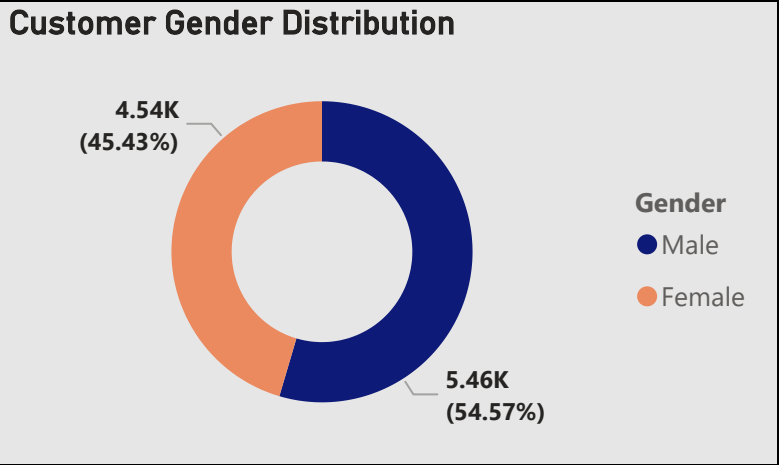
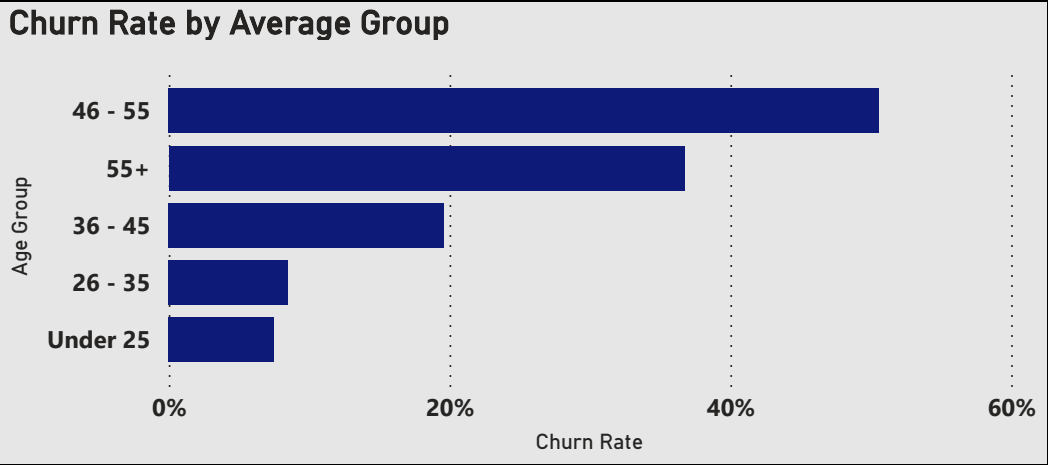
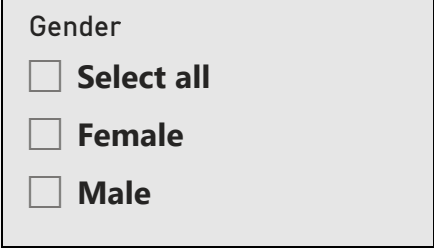
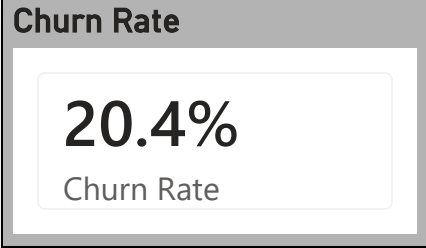
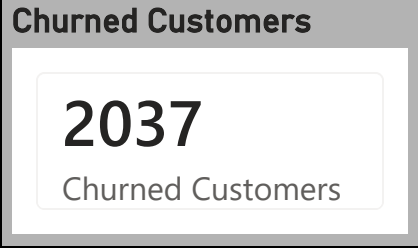
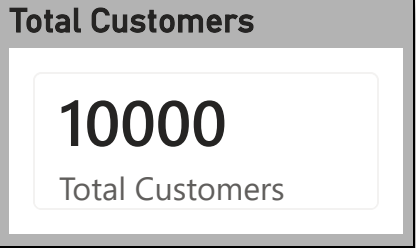
Churn Rate by Country



Churn Rate by Gender



Customer Demographics



Financial Analysis

Total Customers

10000

Total Customers

Churn Rate

20.4%

Churn Rate

Average Balance

\$76,485.89

Avg Balance

Average Credit Score

650.53

Avg Credit Score

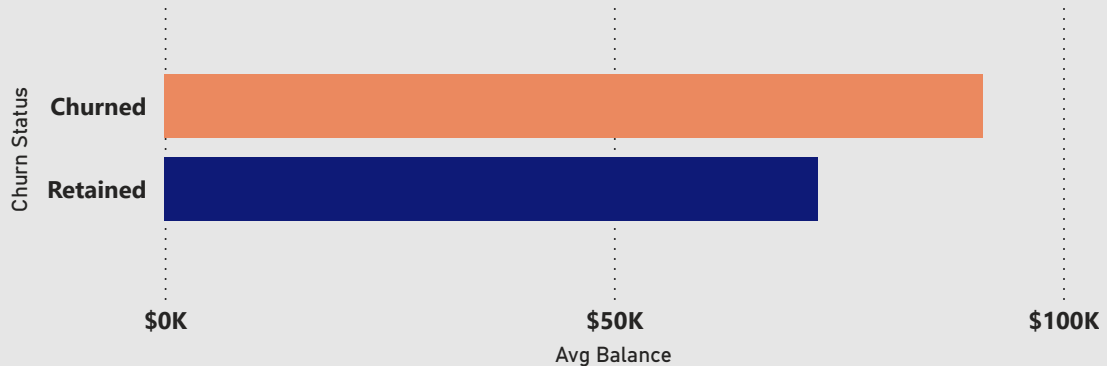
Geography

- ☐ France
- ☐ Germany
- ☐ Spain

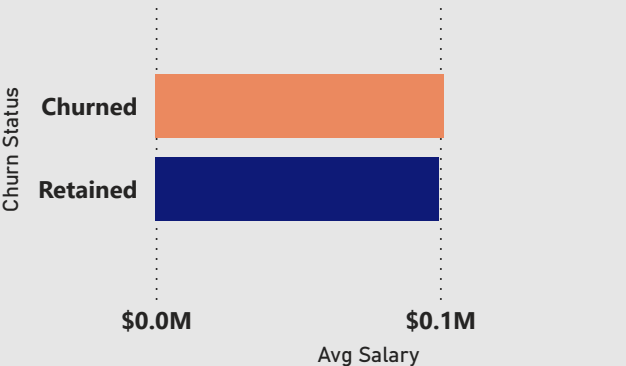
Churn Status

- ☐ Select all
- ☐ Churned
- ☐ Retained

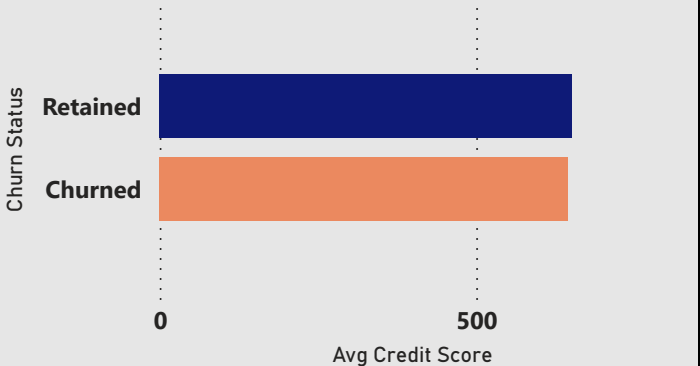
Average Balance - Churned vs Retained



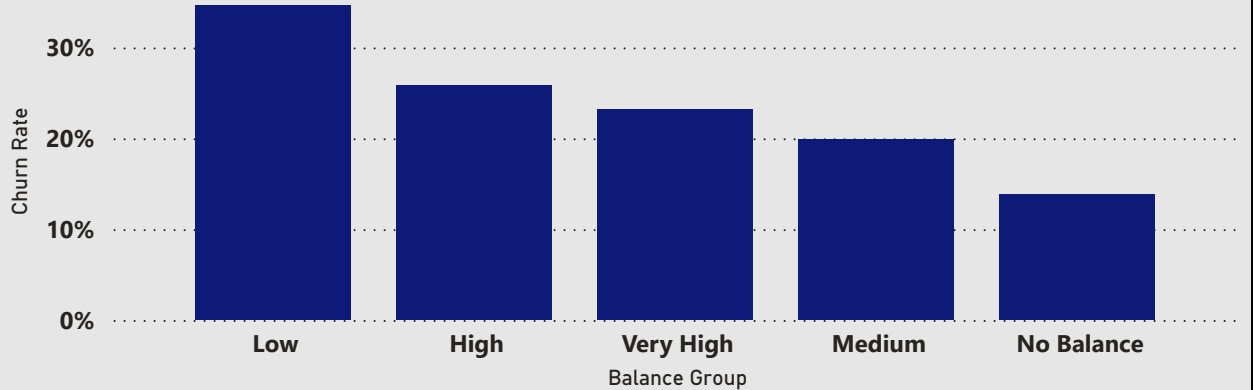
Average Salary - Churned vs Retained



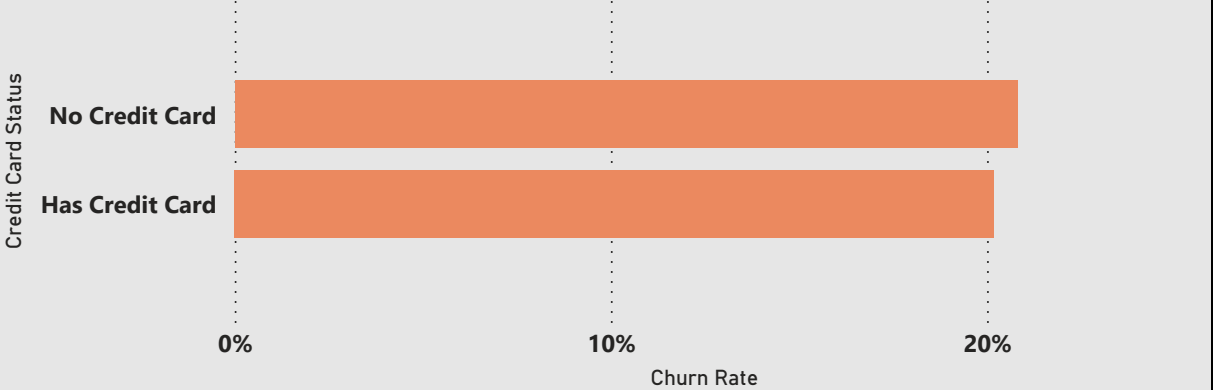
Average Credit Score - Churned vs Retained



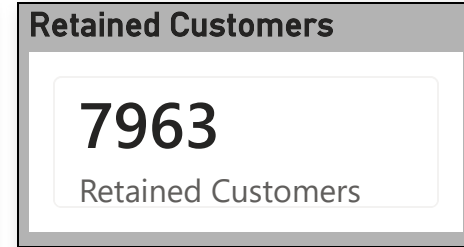
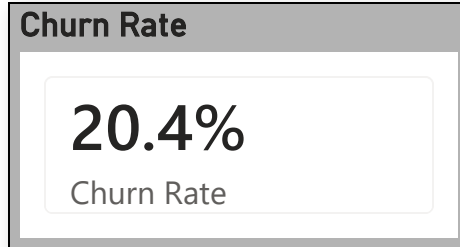
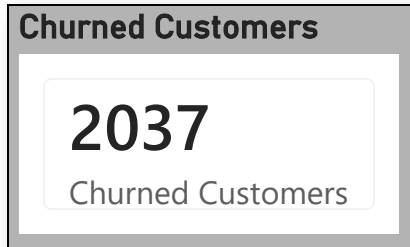
Churn Rate by Balance Group



Churn Rate by Credit Card Status



Model Insights



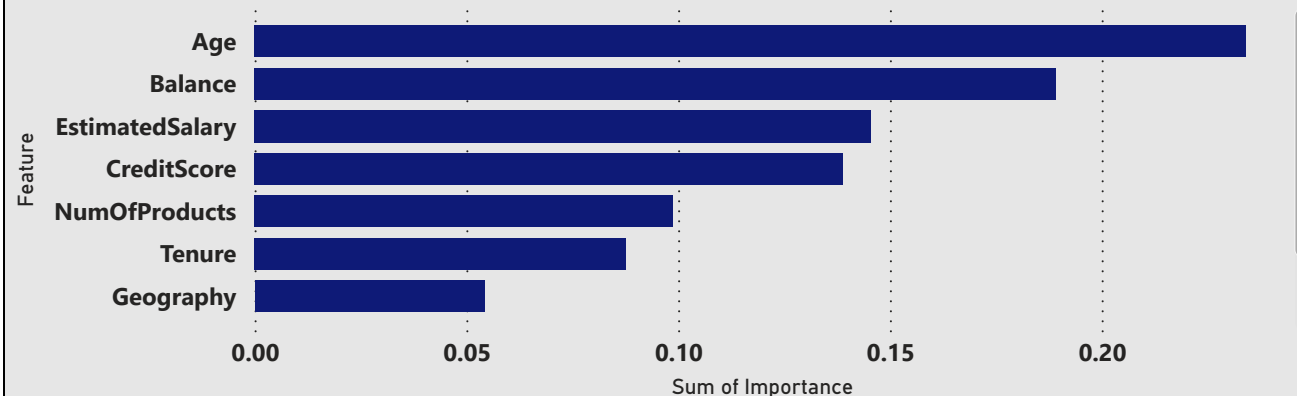
Geography

☐ France

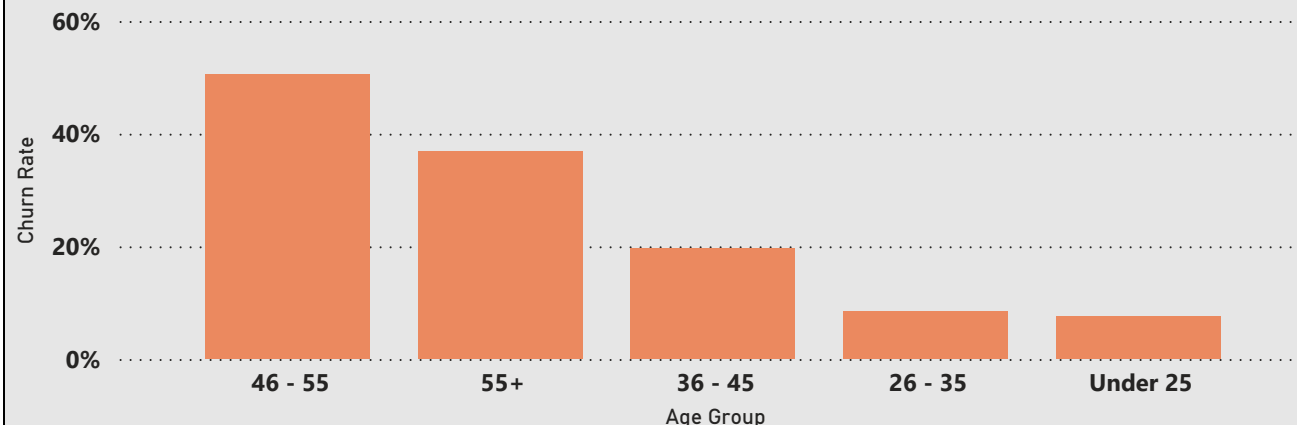
☐ Germany

☐ Spain

Top Churn Prediction Factors - Random Forest Model



Churn Risk by Age Group



MODEL PERFORMANCE SUMMARY

Best Model: Random Forest

Logistic Regression:

- Accuracy: 81.10%
- ROC AUC Score: 0.7789

Random Forest:

- Accuracy: 86.60%
- ROC AUC Score: 0.8576

The Random Forest model correctly predicts 9 out of 10 customers.

KEY FINDINGS & RECOMMENDATIONS

● High Risk Customers:

- Aged between 40 and 55
- Located in Germany
- Have only 1 product
- Are inactive members
- Have low account balances

✓ Recommended Actions:

- Target inactive members with re-engagement campaigns
- Offer product bundles to single product customers
- Develop Germany-specific retention strategies
- Create special programs for middle-aged customers
- Flag high balance customers showing inactivity signs